HOUSE PRICE PREDICTION

TEAM VORTEX

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PROBLEM STATEMENT

> House prices are influenced by multiple factors, making accurate prediction challenging. This project develops a reliable model to estimate prices based on key housing features.

OBJECTIVE

To predict **median house value** using given features

To identify important factors affecting **house prices**

To apply **ANOVA** for statistical analysis

DATASET

kaggle

FEATURES USED

- longitude
- latitude
- housing_median_age
- total_rooms
- total_bedrooms
- population
- households
- median_income
- median_house_value

METHODOLOGY

Data Cleaning and Preprocessing

ANOVA Test

Model Building

Model Evaluation

EDA

MODELS USED

Linear Regression

- Used as a baseline model
- Assumes linear relationship

Random Forest Regression

- Handles non-linear relationships
- Provides Better Accuracy

CONCLUSION

> This project combines statistical analysis and machine learning to predict house prices. ANOVA helps identify significant variables, and regression models are used to make accurate predictions.

REFERENCES

Emad, Y. (2026). House price prediction dataset. Kaggle.
https://www.kaggle.com/datasets/youssefemad004/house-price-prediction